CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 1: Rational Numbers

Topic-wise short notes with solved examples

Rational numbers, their standard form, the four operations and their properties, representation on the number line, and inserting rational numbers between two given ones.

1 What a rational number is

- A **rational number** is a number of the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.
- Every natural number, whole number, integer and fraction is a rational number: $5 = \frac{5}{1}$, $0 = \frac{0}{1}$, $-3 = \frac{-3}{1}$.
- A rational number is **positive** if p and q have the same sign, and **negative** if they have opposite signs.
- **Equivalent rational numbers**: multiplying or dividing both terms by the same non-zero integer gives an equal number. $\frac{2}{3} = \frac{4}{6} = \frac{-6}{-9}$.
- **Standard form**: the denominator is positive and the two terms have no common factor other than 1.

Example 1.1 Write $\frac{36}{-48}$ in standard form.

SOLUTION

Divide both terms by their HCF 12 and make the denominator positive.

$\frac{-3}{4}$

2 Comparing rational numbers

- Write both numbers with positive denominators. Convert to a common denominator (the LCM) and compare the numerators.
- Or cross-multiply: for positive denominators, $\frac{a}{b} > \frac{c}{d}$ exactly when $ad > bc$.
- Every positive rational number is greater than 0, and every negative rational number is less than 0.

Example 2.1 Which is greater: $\frac{-5}{6}$ or $\frac{-7}{9}$?

SOLUTION

LCM of 6 and 9 is 18: $\frac{-15}{18}$ and $\frac{-14}{18}$.

$\frac{-7}{9}$ is greater

Example 2.2 Arrange in ascending order: $\frac{2}{3}, \frac{-1}{2}, \frac{5}{6}, \frac{-3}{4}$.

SOLUTION

With denominator 12: $\frac{8}{12}, \frac{-6}{12}, \frac{10}{12}, \frac{-9}{12}$

$\frac{-3}{4}, \frac{-1}{2}, \frac{2}{3}, \frac{5}{6}$

3 The four operations

- **Addition**: same denominators, add the numerators. Different denominators, first convert to the LCM.
- **Subtraction**: add the additive inverse. $\frac{a}{b} - \frac{c}{d} = \frac{a}{b} + \left(\frac{-c}{d}\right)$.
- **Multiplication**: $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$.
- **Division**: multiply by the reciprocal. $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$, where $c \neq 0$.

Example 3.1 Find $\frac{-3}{8} + \frac{5}{12}$.

SOLUTION

LCM of 8 and 12 is 24: $\frac{-9}{24} + \frac{10}{24}$

$\frac{1}{24}$

Example 3.2 Find $\frac{7}{15} - \frac{-3}{10}$.

SOLUTION

$\frac{14}{30} + \frac{9}{30}$

$\frac{23}{30}$

Example 3.3 Find $\frac{-4}{9} \times \frac{15}{-16}$ and $\frac{-8}{21} \div \frac{4}{7}$.

SOLUTION

$\frac{-60}{-144} = \frac{5}{12}$ and $\frac{-8}{21} \times \frac{7}{4} = \frac{-2}{3}$

$\frac{5}{12}$ and $\frac{-2}{3}$

4 Properties

Property	Addition	Subtraction	Multiplication	Division
Closure	Yes	Yes	Yes	Yes, except division by 0
Commutative	Yes	No	Yes	No
Associative	Yes	No	Yes	No

- **Additive identity**: $\frac{a}{b} + 0 = \frac{a}{b}$. **Multiplicative identity**: $\frac{a}{b} \times 1 = \frac{a}{b}$.
- **Additive inverse** (negative) of $\frac{a}{b}$ is $\frac{-a}{b}$: their sum is 0.
- **Multiplicative inverse** (reciprocal) of $\frac{a}{b}$ is $\frac{b}{a}$, where $a \neq 0$: their product is 1. Zero has no reciprocal.
- **Distributive property**: $\frac{a}{b} \times \left(\frac{c}{d} + \frac{e}{f}\right) = \frac{a}{b} \times \frac{c}{d} + \frac{a}{b} \times \frac{e}{f}$. It also holds over subtraction.
- 1 and -1 are the only rational numbers equal to their own reciprocals.

Example 4.1 Write the additive inverse and the multiplicative inverse of $\frac{-5}{7}$.

SOLUTION

$\frac{5}{7}$ and $\frac{-7}{5}$

Example 4.2 Use the distributive property to find $\frac{3}{5} \times \frac{7}{11} + \frac{3}{5} \times \frac{4}{11}$.

SOLUTION

$\frac{3}{5} \times \left(\frac{7}{11} + \frac{4}{11}\right) = \frac{3}{5} \times 1$

$\frac{3}{5}$

Example 4.3 What should be added to $\frac{-5}{8}$ to get $\frac{3}{4}$?

SOLUTION

$\frac{3}{4} - \left(\frac{-5}{8}\right) = \frac{6}{8} + \frac{5}{8}$

$\frac{11}{8}$

Example 4.4 The product of two rational numbers is $\frac{-14}{27}$. One of them is $\frac{7}{9}$. Find the other.

SOLUTION

$\frac{-14}{27} \div \frac{7}{9} = \frac{-14}{27} \times \frac{9}{7}$

$\frac{-2}{3}$

5 On the number line

- Positive rational numbers lie to the right of 0 and negative ones to the left.
- To mark $\frac{p}{q}$: divide each unit into q equal parts and count p parts from 0 (to the left if negative).
- For an improper fraction, first write it as a mixed number: $\frac{-7}{3} = -2\frac{1}{3}$ lies between -3 and -2.



The points 2/3 and −2/3 on the number line

6 Rational numbers between two rational numbers

- Between any two different rational numbers there are **infinitely many** rational numbers.
- **Mean method**: $\frac{a+b}{2}$ lies between a and b. Repeat to get more.
- **Common denominator method**: write both with the same denominator; if there is not enough room between the numerators, multiply both terms by 10 (or any suitable number).

Example 6.1 Find a rational number between $\frac{1}{4}$ and $\frac{1}{3}$.

SOLUTION

$\frac{1}{2}\left(\frac{1}{4}+\frac{1}{3}\right) = \frac{1}{2} \times \frac{7}{12}$

$\frac{7}{24}$

Example 6.2 Insert four rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$.

SOLUTION

Multiply both terms by 5: $\frac{10}{25}$ and $\frac{15}{25}$.

$\frac{11}{25}, \frac{12}{25}, \frac{13}{25}, \frac{14}{25}$

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7 Common mistakes

- Adding numerators and denominators separately: $\frac{1}{2}+\frac{1}{3}$ is not $\frac{2}{5}$.
- Thinking $\frac{-5}{6}>\frac{-4}{6}$ because 5 is greater than 4. For negatives the order reverses.
- Leaving a negative denominator in the answer.
- Taking the reciprocal as the negative: the reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$, not $\frac{-2}{3}$.
- Inverting the first fraction instead of the second in a division.
- Saying 0 has a reciprocal.

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